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Less pain, same gain? A randomized controlled trial comparing assisted- versus non-assisted Nordic Hamstring exercise training in elite youth football players

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ABSTRACT

Introduction: The Nordic Hamstring exercise (NHE) increases hamstring strength, but muscle soreness and excessive fatigue may hinder adoption. The assisted NHE (A-NHE), with an elastic band around the chest lowering the force in the eccentric lowering phase, may be a less demanding alternative. We investigated the effectiveness of A-NHE versus NHE on hamstring strength, muscle soreness, and perceived exertion.
Methods: We randomized 31 youth elite football players (under-14 and -15) from a single professional football academy to either NHE or A-NHE, who performed the same supervised 8-week program. The A-NHE was performed with an elastic band, progressing from heavy to light assistance. A blinded tester assessed Nordic hamstring strength, while hamstring muscle soreness and perceived exertion were assessed during each session.
Results: Intention-to-treat analysis showed no group by time interaction ($p = 0.61$) for Nordic hamstring strength. Both groups improved significantly (A-NHE mean difference: 3.7 kg, 95% CI [2.1; 5.3]; NHE mean difference: 3.2 kg, 95% CI [1.2; 5.2]) corresponding to approximately 20%. The NHE group reported significantly higher hamstring muscle soreness (beta = 1.77, $p < 0.01$) and perceived exertion (beta = 1.54, $p < 0.01$).
Conclusion: Eight weeks of assisted-versus regular Nordic Hamstring training led to similar strength gains, but with lower hamstring muscle soreness and perceived exertion.

1. Introduction

Hamstring strain injuries are the most prevalent injury type in professional football and may constitute up to 24 % of all injuries (Ekstrand et al., 2023a). While the risk is considered to increase with age, with the early twenties representing a critical point (Timmins et al., 2016), hamstring strain injuries is also a significant issue in youth elite football (Materne et al., 2021). Noteworthy, a rapid increase in the percentage hamstring injuries relative to all injuries seems to occur around the under-14 level (from approx. 3–5 % at under-13 and below to ~10 % at under-14 to under-18) (Materne et al., 2021) co-inciting with an increase in on-field time (Light et al., 2021).

In the last two decades there has been a slight gradual increase in both injury incidence and burden of hamstring strain injuries in the top European football clubs (Ekstrand et al., 2023a). Several factors may predispose athletes to hamstring strain injuries with poor hamstring strength and short biceps femoris fascicle length commonly mentioned as modifiable risk factors (Timmins et al., 2016). Although the literature on poor hamstring strength as a risk factor is inconsistent (Opar et al., 2021), a recent Delphi survey among practitioners in the Big-5 European leagues found consensus for an eccentric strength focus as a strategy to mitigate risk of muscle injuries (McCall et al., 2020), with similar findings in UK youth football academies (Read et al., 2018).

The Nordic Hamstring exercise is a high-intensity eccentric exercise (Van Hooren et al., 2022) that offers an effective way to increase both eccentric hamstring strength and biceps femoris long head fascicle

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length (Bourne et al., 2017; Ishoi et al., 2018). The preventive effectiveness of the NHE among male football players have been established in large randomized controlled trials with a hamstring injury risk reduction of up to 65% (Ishøi et al., 2020). Similar risk reductions have been noted in both women's and men's professional football when implemented team-wide compared to teams using an individualized implementation (Ekstrand et al., 2022, 2023b). The Nordic Hamstring exercise may also result in performance gains with improvements in sprint acceleration (Ishoi et al., 2018), change of direction (Siddle et al., 2019), and counter movement jump (Bright et al., 2023) performance documented across different sports and age groups.

Recognizing the importance of effective injury prevention (Larruskain et al., 2022) and physical development strategies (Dugdale et al., 2021) in youth football players to improve chances of progression towards senior football, eccentric training is considered important (Bright et al., 2023), and may be relevant to implement already at the under-14 level due to the rapid rise in hamstring injury risk (Materne et al., 2021). However, despite the positive attributes (Bright et al., 2023), the implementation of the Nordic Hamstring exercise remains a challenge (Bahr et al., 2015; Ekstrand et al., 2022, 2023b). The barriers towards adoption often include hamstring muscle soreness, players negative perceptions of performing the exercise (Chesterton et al., 2021, 2022), and difficulties with programming into the micro cycle during periods with ≤4 days between matches (McCall et al., 2020). Acknowledging these challenges, the assisted Nordic Hamstring exercise has been proposed as an alternative approach (Matthews et al., 2015). This variation introduces an elastic band or cable to reduce the resistance (Matthews et al., 2015) and it results in less perceived exertion and acute fatigue (Burrows et al., 2020), while also acutely reducing knee moments (Alt, Knicker, et al., 2021), biceps femoris muscle activity (Burrows et al., 2020), and knee angular velocities (Alt, Knicker, et al., 2021), potentially resulting in less muscle soreness. Indeed, hamstring muscle soreness can be modulated by varying the training dosage of the Nordic Hamstring exercise, indicating that it is not an inherent adverse effect but rather a function of programming (Behan et al., 2023). Furthermore, the assisted Nordic Hamstring exercise increases time under tension and knee range of motion (Alt, Knicker, et al., 2021) which may drive muscular adaptions (Toigo & Boutellier, 2006). A few studies have included the assisted Nordic Hamstring exercise in interventions (Alt et al., 2018, 2021b), however, to the best of our knowledge the assisted Nordic Hamstring exercise has never studied longitudinally in isolation.

We aimed to compare the effectiveness of the assisted Nordic Hamstring exercise to the conventional Nordic Hamstring exercise on eccentric hamstring strength, hamstring muscle soreness, and perceived exertion in elite youth football players.

2. Materials and methods

2.1. Trial design

This is the primary trial report of a single assessor-blinded randomized controlled trial with a 2-group parallel design. The intervention period was an 8-week NHE intervention based on the van der Horst protocol (van der Horst et al., 2015) performed as either assisted using elastic bands (assisted NHE) or non-assisted (non-assisted NHE). The outcome measures included the Nordic hamstring strength measured using a commercial Nordic hamstring testing device (H-Station, Fysiometer, Aaborg, Denmark) pre- and post-intervention, hamstring muscle soreness measured before each session using a 0–10 visual analog scale, and perceived exertion measured after each session using the Borg CR10 scale (Foster et al., 2001). The reporting adheres to the Consolidated Standards of Reporting Trials (CONSORT) and ethical approval was obtained from the Danish Ethics Committee of the Capital Region as part of a larger project (ID: H-19026064). Due to the scarcity in the literature concerning longitudinal adaptations of the assisted Nordic Hamstring exercise, the study was not preregistered as we had no clear predefined primary outcome measure and hypotheses, which is requested by www.clinicaltrials.gov. Instead, we adopted a flat outcome structure, and thus consider the trial to be exploratory.

2.2. Study settings

All data was collected at the facilities of a professional 1st tier football club (Danish Superliga) from September to November 2021. The study was conducted during the in-season period when the players typically had five football training sessions (approx. 8–10 h of training) and 1–2 matches per week. In addition, gym-based training was performed twice weekly, mainly consisting of compound exercises, such as squat and deadlift, Copenhagen Adduction, plyometrics, and upper body work. Gym sessions were conducted as normal, with no additional eccentric-biased hamstring training besides the interventions being performed. The gym sessions were typically performed prior to on-field training sessions between matchday +1 and −2.

2.3. Participants

Thirty-one elite youth football players from the under-14 and -15 squads from a single professional youth football academy area were included through convenience sampling (Fig. 1). Exclusion criteria were any hamstring pain during baseline or follow-up testing reported by the player to influence testing performance; inability to participate fully in the regular training schedule at the time of baseline testing and/or follow-up testing; or any hamstring injury during the intervention period. Information on the study was distributed to parents, players, and relevant stakeholders at the club prior to study initiation. All parties provided consent on the execution of the study.

2.4. Intervention

Both intervention groups performed an 8-week progressive eccentric strengthening program for the hamstring muscles using the Nordic Hamstring exercise based on the initial 8 weeks of the van der Horst protocol (Table 1). (van der Horst et al., 2015)

The Nordic Hamstring exercise is originally described as a partner-assisted exercise, performed without additional equipment or external load (Mjolsnes et al., 2004). In a kneeling position (90° of knee flexion), a partner fixates the ankles to ensure constant contact between the lower legs and the ground throughout the movement. Subsequently, the person leans forward as slowly as possible, while attempting to resist the forward motion by maximally activating the hamstring muscles in the eccentric phase. Participants were asked to maintain a straight torso and slightly flexed hips during the whole range of motion. Participants were asked to use their hands to buffer the fall after they "let go" and to forcefully push themselves back to the starting position with their hands to minimize hamstring force production in the corresponding concentric contraction phase (Mjolsnes et al., 2004). Players allocated to the Nordic Hamstring group performed the exercise as described above, while players allocated to the assisted Nordic Hamstring group performed the exercise using an elastic band to assist the downward movement (Matthews et al., 2015). The elastic band was applied around the chest of the player to lower the force exerted by the upper body, in a progressive semi-structured manner from heavy to light resistance. Three different resistances of elastic bands (black; heavy resistance, green; intermediate resistance, yellow; light resistance; Theraband, Akron, Ohio, USA) with standardized lengths mounted to the wall were used for assistance. The length change of the elastic bands during the execution of the exercise was approximately 100–150 % compared to resting length. All players started with the black, progressing to the green, and finally the yellow elastic band. Players had to use each elastic band for a minimum of 2 weeks (4 training sessions). In the last training session of a phase, the execution of the exercise was evaluated during the training

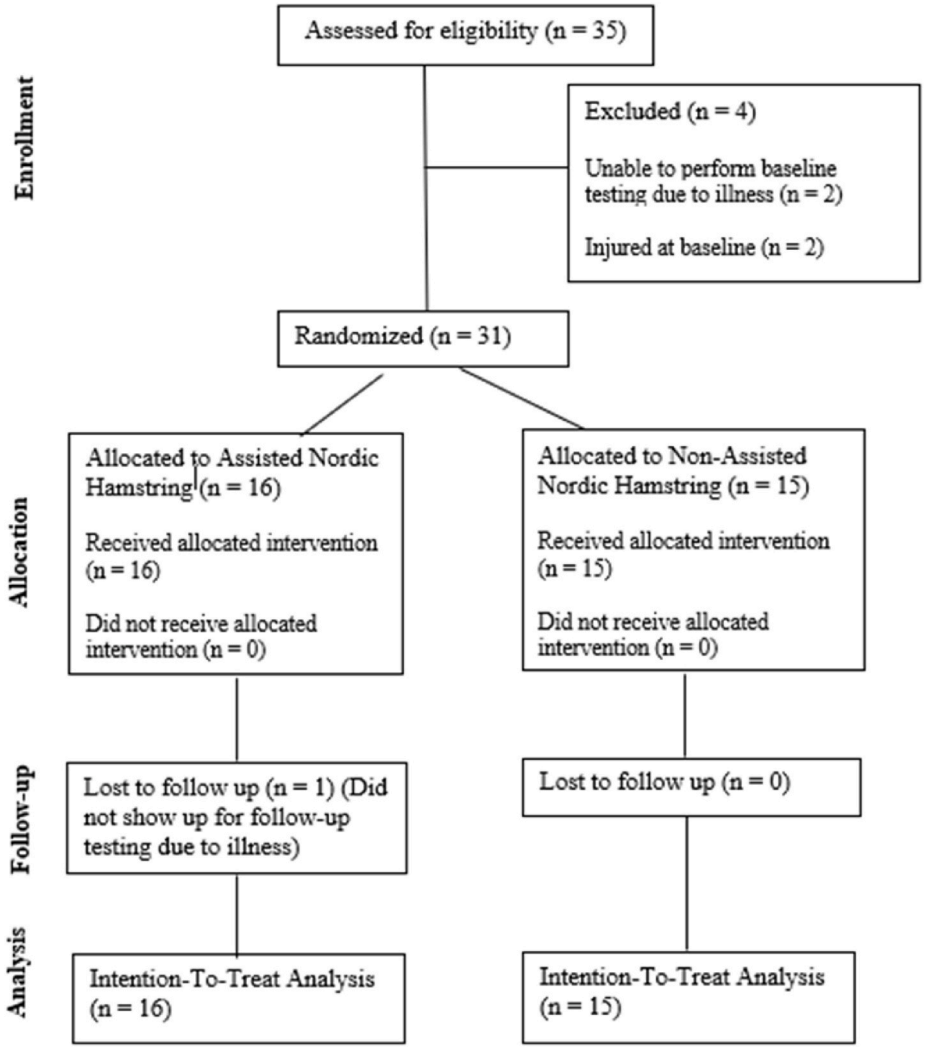

Fig. 1. Flow of participants.

Table 1
The initial 8-weeks of the Van der Horst Nordic Hamstring Protocol.

Week	Training sessions per week	No. of Sets per Training Session	Repetitions per Set
1	1	2	5
2	2	2	6
3	2	3	6
4	2	3	6, 7, 8
5	2	3	8, 9, 10
6–8	2	3	10, 9, 8

session, by the physiotherapists who supervised the exercise, according to pre-defined criteria: the player being able to lower the upper body until reaching the ground with a slow consistent angular velocity. Players who fulfilled the criteria, progressed to the lighter resistance for at least two weeks, while players who did not fulfill the criteria continued with the current resistance before being re-evaluated in the final training session of the subsequent week. Players were allowed to progress to no resistance following the use of the yellow elastic band, and this could occur at earliest in week 7, however, no players achieved this (Fig. 2).

Players were instructed to alternate between performing one set and manually fixating the ankles of the partner player, respectively, giving an inter-set recovery time of 90–120 s. All players were familiar with Nordic Hamstring exercise training, however, with no recent history of structured use. Recovery time between exercise sessions was planned to be at least 48 h with sessions performed between matchday +1 and matchday −2. All NHE training was performed as group training sessions after regular football training sessions to avoid exercise-induced hamstring muscle fatigue that may interfere with the regular football training. All sessions were supervised by two physiotherapists, who also registered training compliance as completed number of sessions.

2.5. Outcome measure

We used a flat outcome structure (no outcome hierarchy) investigating the between-group difference in Nordic Hamstring strength, hamstring muscle soreness, and perceived exertion. The Nordic Hamstring strength was measured using a commercial Nordic Hamstring testing device with a sampling rate of 100 Hz (H-Station, Fysiometer, Aalborg, Denmark) (Intraclass correlation coefficient: 0.96, Standard Error of Measurement: 6.7 %) (Støve et al.). The force was transferred via Bluetooth to the FysioMeter software on a tablet, which applies a 4th-order Butterworth lowpass filter with a cutoff frequency of 20 Hz before calculating peak force. Participants were kneeling on a padded board with heels fastened against two Wii boards (Nintendo, Kyoto, Japan) used for sampling force. From this position, participants performed the Nordic Hamstring exercise. Three warm-up repetitions followed by a minimum of three repetitions were performed. A trial was deemed acceptable when the force output reached a peak force value,

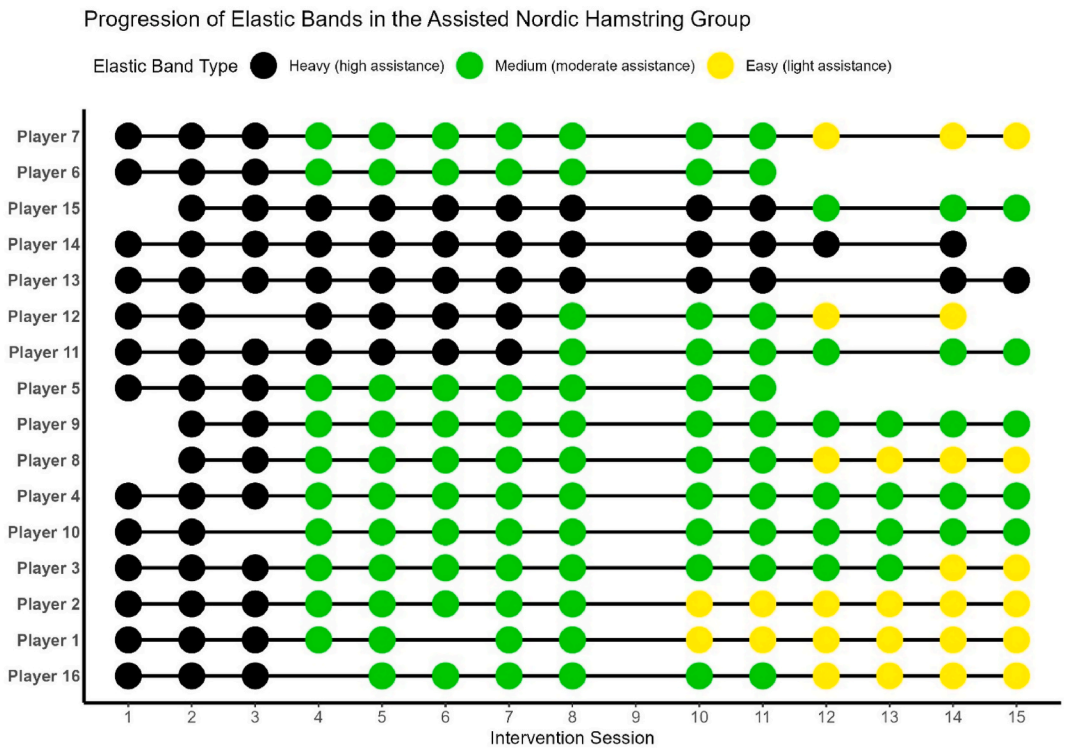

Fig. 2. Progression of elastic band resistance used by each player (y-axis) for each training session (x-axis).

followed by a rapid decline in force, and was performed as slowly as possible. This was emphasized before and during each repetition by via verbal feedback from the tester as recommended (Alt & Schmidt, 2021). Peak strength was calculated as the maximum mean peak force of both legs (Opar et al., 2013). If the player was able to perform the Nordic Hamstring exercise without "letting go" (i.e. consistent slow movement to the ground) additional external weight was supplied in 5-kg incremental, until maximum force was reached (Bourne et al., 2017). No player performed more than 5 repetitions. All players were familiar with strength testing method as this was part of an assessment procedure performed quarterly. Hamstring muscle soreness was collected prior to each session, where participants were asked to fill in mean hamstring muscle soreness since the previous session on a Visual Analog Scale (0-10) using pen and paper. Pain ratings using VAS have been widely used to track delayed onset muscle soreness showing strong agreement with more descriptive Likert Scales (Impellizzeri & Maffiuletti, 2007). Following the session, participants were asked to rate the perceived exertion of the hamstring muscles using the Borg CR10 method proposed by Foster et al. (Foster et al., 2001)

2.6. Randomization and blinding

Following baseline testing, a person not involved in the study conducted the randomization with an allocation ratio of 1:1 and stratified by team to accommodate factors such as maturity status and training schedule using an online randomization tool (Random List, https://www.randomlists.com/team-generator). The person who conducted the baseline and follow-up testing and statistical analyses was blinded to group allocation.

2.7. Sample size considerations

While the sample size was based on convenience, our inclusion of 31 players, and a power and alpha-level of 0.8 and 0.05, respectively, allowed us to detect a between group difference of a medium to large effect size corresponding to cohen's *d* of 1 (calculated using the *pwrss.z.2means* function from the *pwrss* package in R). We considered a medium to large effect size to be appropriate due to the explorative nature of the study.

2.8. Statistical methods

Analyses were conducted using the R Statistical language (version 4.3.2; R Core Team, 2023) using the following packages: lme4 (version 1.1.35.1) (Bates et al., 2015), lmerTest (version 3.1.3) (Kuznetsova et al., 2017), Matrix (version 1.6.5) (Bates, Maechler, & Jagan, 2024), and rstatix (version 0.7.2) (Kassambara, 2023). The analysis of the Nordic Hamstring strength was performed as intention-to-treat including all players regardless of compliance and follow-up status. Between-group differences in peak Nordic hamstring strength was assessed using a linear mixed model (estimated using nloptwrap optimizer) with group, time, and group by time interaction as fixed effects, and subject as random effect using the *lmerTest::lmer* function. Additionally, the simple main effect of time in each group was analyzed by paired t-tests using the rstatix*t_test* function. Prior to analyses, the data were tested for normality using the Shapiro Wilk test and for Homogeneity of Variances using the Levene's test. All assumptions were met. To investigate between-group differences in rating of perceived exertion (Borg CR10) and hamstring muscle soreness, we fitted linear mixed models (estimated using nloptwrap optimizer) with group, training session, and group by training session interaction as fixed effects, and training session and subject as random effects using the *lmerTest::lmer* function.

3. Results

3.1. Participants

We included 31 players (mean age ± standard deviation (A-NHE vs NHE): 13.3 ± 0.8 vs 13.6 ± 0.5, mean height: 165.5 ± 10 vs 162.9 ± 6.5, mean weight: 46.3 ± 5.5 vs 48.4 ± 8.5), with one player in the assisted NHE group missing follow-up testing because of illness. The

number of completed training sessions were similar between the groups (mean [range] compliance: assisted NHE; 76 % [60–93%], non-assisted NHE; 82 % [66–93%]). The two reasons for missing a training session were 1) inability of the team to schedule the training session due to training/match congestion (this happened twice once for each squad), and 2) illness.

3.2. Nordic Hamstring Strength

There was no significant group by time interaction on Nordic Hamstring strength, indicating no between-group difference in change (between-group difference: −0.57 kg, 95% CI [−2.83; 1.68], p = 0.61). The assisted NHE group improved from 16.5 (3.9) to 20.2 (6.0) kg (mean difference: +3.7 kg (22 %), 95% CI [2.1; 5.3]), whereas the NHE group improved from 16.0 (3.4) to 19.2 (5.1) kg (mean difference: +3.2 kg (20 %), 95% CI [1.2; 5.2]) (Fig. 3).

3.3. Hamstring muscle soreness and perceived exertion

For hamstring muscle soreness, no significant interaction was observed, however, on average across sessions the effect of the NHE group was significant and positive (beta = 1.77, 95% CI [0.66, 2.89], p = 0.002) (Fig. 4).

For the rating of perceived exertion, a significant interaction was observed with the NHE group reporting gradually higher perceived exertion across the intervention (beta = 0.1, 95% CI [0.03; 0.16], p = 0.005). In addition, on average across sessions the effect of the NHE group was significant and positive (beta = 1.54, 95% CI [0.88, 2.20], p < 0.001) (Fig. 5).

4. Discussion

The present study aimed to compare the effectiveness of the assisted- versus non-assisted Nordic Hamstring exercise on eccentric hamstring strength, hamstring muscle soreness, and perceived exertion in elite youth football players. We found no significant between-group difference in Nordic Hamstring strength gains after an 8-week training period in youth elite football players. Both groups demonstrated substantial improvements in Nordic hamstring strength corresponding to ~20%. In addition, lower hamstring muscle soreness and rating of perceived exertion were reported by players in the assisted Nordic Hamstring group. Given the barriers towards implementation of the Nordic Hamstring exercise often pertains to fear of muscle soreness and excessive fatigue (Chesterton et al., 2022), our preliminary findings suggest that the assisted Nordic Hamstring exercise may offer a feasible and potentially more tolerable approach to eccentric hamstring training. This may have the potential to improve adoption and compliance rates that are essential for obtaining a risk reduction effect (Goode et al., 2015). Since we adopted a flat outcome structure with no preregistered primary outcome measure, future studies are needed to verify our results. However, for studies aiming to show superiority in relation to eccentric hamstring strength, the between-group difference of 0.54 kg corresponding to a trivial effect size (Cohen's *d* of 0.14) shown in the present study, may suggest that a sample size large enough to detect at least a small effect size of 0.2 (for the different to be of relevance) equivalent to 394 participants per group are needed.

4.1. Comparison to existing literature

The strength gains observed in the present study overall aligns with previous findings on Nordic Hamstring training in youth athletes (Drury et al., 2020; Freeman et al., 2019; Lacome et al., 2020). Drury et al. (Drury et al., 2020) observed a 19 % and 10 % increase in eccentric hamstring strength following six weeks of twice weekly Nordic Hamstring training in football players pre- and post-peak height velocity, respectively (Drury et al., 2020), whereas Lacome et al. (Lacome et al., 2020) observed ~10 % increase in elite U19 academy football players following either a high- or low-volume protocol. The higher strength gains observed in the present study compared to Lacome et al. (Lacome et al., 2020) may be attributed to 1) the younger population in the present study (U14-U15 versus U19) coupled with only unstructured previous experience with the exercise potentially having more room for improvement (Drury et al., 2020), and/or a higher training volume in the present study (weekly average number of repetitions: 41 across 8 weeks versus 16 repetitions across 6 weeks in the high-volume group (Lacome et al., 2020)). In contrast, Siddle et al. (Siddle et al., 2024)

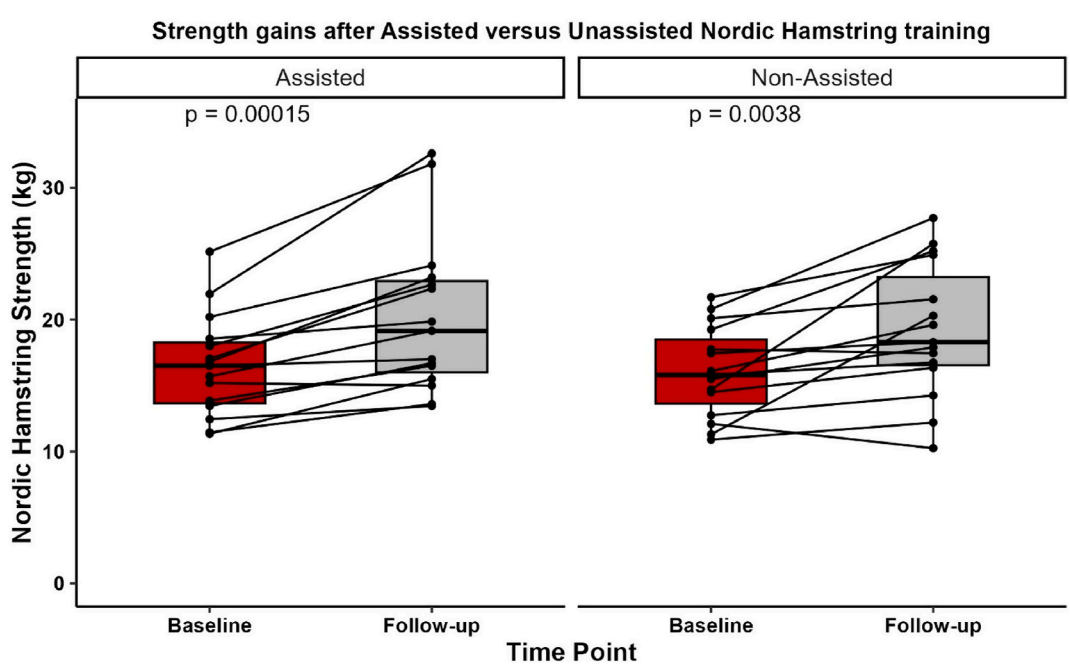

Fig. 3. Changes in the Nordic Hamstring Strength from baseline to follow-up. Box plots show group changes, while lines show individual changes. P-values represent within-group changes. The p-value for the group by time interaction analysis is non-significant at 0.65.

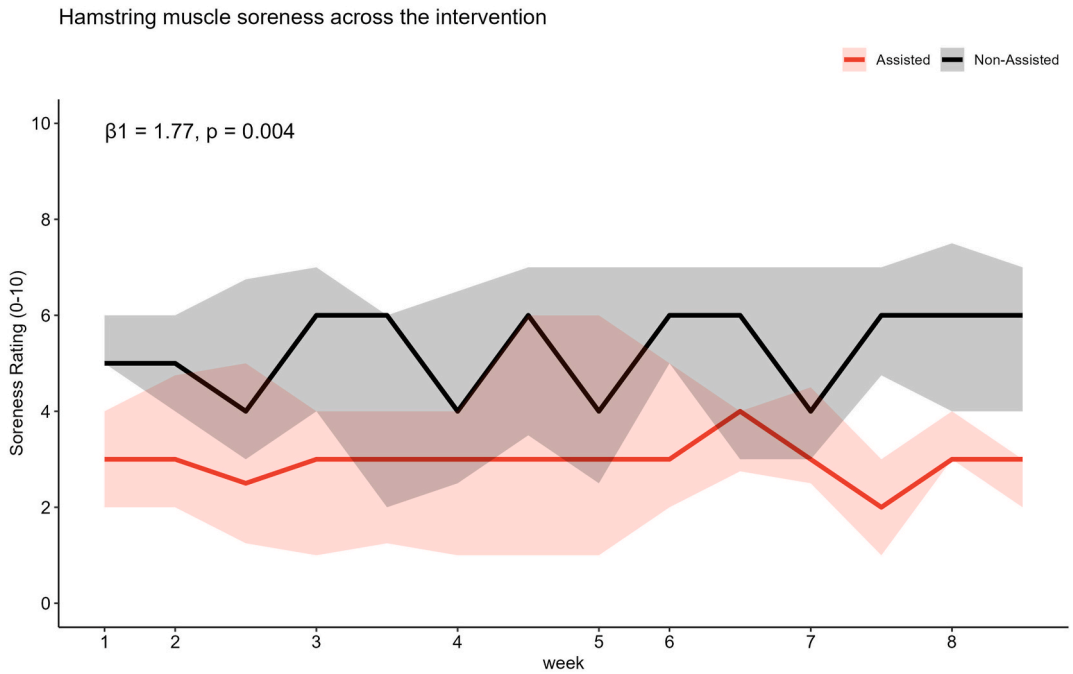

Fig. 4. Median hamstring soreness across the intervention period. The shaded areas represent the interquartile range. Note that only one training session was completed in week 5 due to schedule changes.

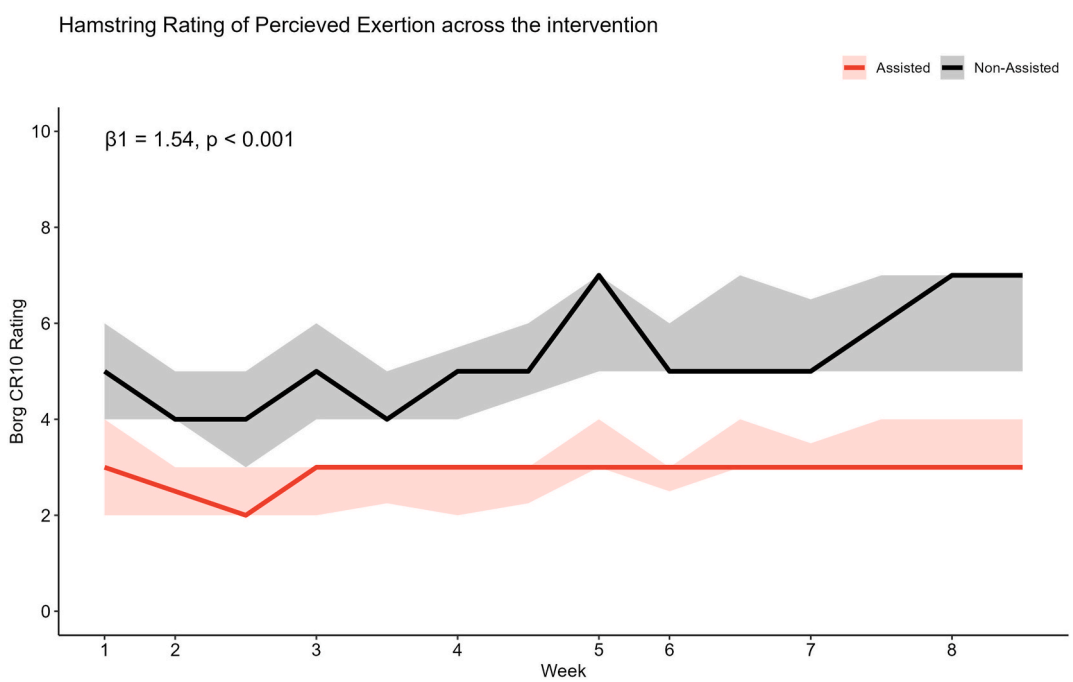

Fig. 5. Median rating of perceived exertion (Borg CR10) across the intervention period. The shaded areas represent the interquartile range. Note that only one training session was completed in week 5 due to schedule changes.

found no improvements in eccentric hamstring strength following a low-volume Nordic Hamstring intervention in elite youth football players.

4.2. Exercise characteristics

Although the assisted Nordic Hamstring exercise is not a new concept (Matthews et al., 2015), surprisingly few studies have been published on the topic (Alt et al., 2018, 2021a; Burrows et al., 2020). Two cross-sectional studies have investigated the exercise characteristics (Alt, Knicker, et al., 2021; Burrows et al., 2020), whereas a single study investigated adaptations following 4-weeks of combined conventional and assisted Nordic Hamstring training in elite male sprinters (Alt et al., 2018). Compared to the conventional Nordic Hamstring exercise, the assisted variation is associated with higher time under tension, greater knee range of motion towards full extension, and lower mean angular velocity (Alt, Knicker, et al., 2021). These parameters are considered relevant determinants for muscle hypertrophy and strength

development (Toigo & Boutellier, 2006), however, they come at an expense of lower peak moment (Alt, Knicker, et al., 2021) and less hamstring muscle activity (Burrows et al., 2020). Although we did not assess exercise characteristics in the present study, and thus can only assume that these discrepancies exist in our cohort, the lower muscle soreness and perceived exertion in the assisted Nordic Hamstring group may indeed suggest less eccentric loading. Regardless, the present study extends on the cross-sectional study findings on exercise characteristics (Alt, Knicker, et al., 2021; Burrows et al., 2020) by providing the first longitudinal data on how the assisted versus non-assisted Nordic Hamstring exercise influence eccentric strength gains, muscle soreness, and perceived exertion in a cohort of elite youth football players unaccustomed to heavy eccentric hamstring training. We can only speculate on potential reasons for the similar strength gains observed between the groups. The conventional Nordic Hamstring exercise is, for many athletes, associated with supramaximal (>1 repetition maximum) loading (Alt, Knicker, et al., 2021), which may drive superior eccentric strength gains compared to either 1) traditional strength training where the execution of the eccentric phase is limited by the concentric strength or 2) eccentric-only training with lighter loads, as would be assumed to be the case for the assisted Nordic Hamstring exercise (Douglas et al., 2016). However, this nature of the exercise may also force athletes, especially if unaccustomed to heavy eccentric loading, to "let go" early in the eccentric lowering phase reducing peak torque. Contrary, the assisted Nordic Hamstring exercise is characterized by lower eccentric muscle forces and activation, and slower angular velocities increasing time under tension, impulse, and greater knee range of motion (Alt, Knicker, et al., 2021). Combined these characteristics may have beneficial implications for athletes who are either unable to "slow" down the downward eccentric phase or "let go" early, since slower angular velocities may be preferred during eccentric strength training (Pereira et al., 2016). In addition, greater knee range of motion towards end range in the assisted Nordic Hamstring exercise theoretically increases hamstring muscle activity (Hegyi et al., 2018), fascicle lengthening, and muscles forces (Van Hooren et al., 2022). Apart from differences in exercise characteristics, the similar strength gains may also simply be explained in the context of the included population representing players with relatively little experience with heavy eccentric hamstring training. Noteworthy, no players in the assisted Nordic Hamstring group progressed to no assistance, and most players maintained the heavy or medium resistance elastic band throughout the intervention. While progression to a lower resistance elastic band or even no band was only allowed when the player showed a slow consistent angular velocity during the downward eccentric phase, this suggests that the assisted Nordic Hamstring exercise still provided a heavy eccentric stimulus.

4.3. Clinical implications

Eccentric hamstring strength training is an important part of long-term athletic development (Bright et al., 2023), and is considered important to mitigate the risk of hamstring strain injuries across youth (Read et al., 2018) and senior elite football (McCall et al., 2020) and to induce favorable muscle- and tendinous adaptations (Bright et al., 2023; Douglas et al., 2016). The Nordic Hamstring exercise reduces the risk of hamstring strain injury in football players (Ishøi et al., 2020), while also enhancing performance metrics such as sprint acceleration (Ishoi et al., 2018) and change of direction (Siddle et al., 2024). Implementation of the Nordic Hamstring exercise may be relevant already from the under-14 level, where a rapid increase in the proportion of hamstring increased seem to occur which further gradually increases with age (Materne et al., 2021). However, implementation remains a challenge (Bahr et al., 2015; Ekstrand et al., 2022, 2023b), with hamstring muscle soreness and excessive fatigue often cited as reasons for low compliance (Chesterton et al., 2021, 2022). The present study presents preliminary evidence that the assisted Nordic Hamstring variation may offer a feasible and potential more tolerable approach to eccentric hamstring training in settings where hamstring muscle soreness and negative perception of the exercise may be barriers towards implementation (Chesterton et al., 2022), which may be the case in youth or senior players unaccustomed to heavy eccentric hamstring training. This may have important clinical implications as the risk reduction effect associated with the Nordic Hamstring exercise seems to be influenced by compliance rates (Goode et al., 2015). In the context of physical development of youth football players (Bright et al., 2023; Read et al., 2018) successful implementation of structured eccentric hamstring training during the academy years may facilitate positive attitudes towards the exercise and potentially increase the chances of players adopting the exercise when progressing to the senior level although this needs to be explored in further details. During the academy years, implementation of the Nordic Hamstring exercise may also help lower the risk of time-loss from football training due to hamstring injuries, which has been identified as a factor that may hamper progression towards the first team level (Larruskain et al., 2022).

4.4. Methodological considerations

In the assisted Nordic Hamstring protocol, all players used the same elastic band for two weeks before individually progressing towards lighter elastic bands, and this was based on feasibility. During the training sessions, we adjusted the height of the fixation point of the elastic band based on the height of the player, but we cannot exclude that further individualization of the elastic bands based on hamstring strength, body mass etc. could have yielded better outcomes. In relation to this, we do not know how much each elastic band affected the force output, and hamstring muscle activation, across players. This has recently been investigated, showing that as little as 5 % assistance of body mass provided by a cable resistance machine was able to reduce the force output (Burrows et al., 2020). Despite these limitations, both muscle soreness and perceived exertion ratings would suggest that the assisted Nordic Hamstring intervention led to lower force outputs during the execution. While strength increases were similar between intervention groups, we do not know whether this translates to similar performance gains and/or preventive effects. Of particular interest, not investigated in the present study, is if the assisted Nordic Hamstring intervention also increases biceps femoris fascicle length as a surrogate measure for its potential preventive effect on hamstring strain injuries (Timmins et al., 2016). During baseline and follow-up testing, additional external weight was added if the player was able to "control" his body to the floor, however a recent study published after the completion of the current study questions the validity of such an approach showing that Nordic Hamstring force may increase with external load regardless of the ability to control the descending movement (Amundsen et al., 2023). The recording of muscle soreness may be due to recall bias. In previous studies, where muscle soreness was tracked in the days following eccentric activities, pain ratings were collected during the execution of a standardized movement (Impellizzeri & Maffiuletti, 2007). This was not done in the present study due to practical reasons, thus more precise monitoring of muscle soreness following the assisted versus non-assisted Nordic Hamstring exercise is warranted in future studies. Due to the explorative nature of our study, we did not pre-register our study, as it does not pertain to a primary outcome only (Richter et al., 2024), as previously mentioned replication is therefore warranted.

5. Conclusion

In this exploratory study, performed in elite youth male football players, no-between group difference was observed in Nordic Hamstring strength after 8-weeks of training with an assisted-versus non-assisted Nordic Hamstring protocol. Both intervention groups improved Nordic hamstring strength by 20%, but training with the assisted Nordic Hamstring exercise resulted in lower hamstring muscle soreness and less perceived exertion. Considering that hamstring muscle soreness and

perceived exertion of the Nordic Hamstring training are often barriers towards team-wide implementation, the assisted Nordic Hamstring exercise may be a relevant alternative.

CRediT authorship contribution statement

Lasse Ishøi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Simon Meincke:** Writing – review & editing, Project administration, Methodology, Investigation. **Anders P. Lund:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Anders Stenholm:** Writing – review & editing, Project administration, Methodology, Investigation. **Matt DeLang:** Writing – review & editing, Conceptualization. **Kasper Thornton:** Writing – review & editing, Methodology, Conceptualization. **Kristian Thorborg:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Ethical approval

Ethical approval was obtained from the Danish Ethics Committee of the Capital Region as part of a larger project (ID: H-19026064).

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Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.ptsp.2025.02.007.

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